

Unit 11, Lesson 2: Evaluating Expressions with Exponents

Benchmark

MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.

Learning Target



I can evaluate expressions that include exponents.

11.2.1 Warm-Up: Error Analysis

Two students made errors on yesterday’s exit ticket. Their work is shown below.

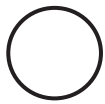
Evaluate $8a + 3b(2a - 4)$ for $a = 2$ and $b = 6$.

Azalea’s Work	Joshua’s Work
$82 + 36(22 - 4)$	$8(2) + 3(6)[2(2) - 4]$
$82 + 36(18)$	$16 + 18(4 - 4)$
$82 + 648$	$16 + 18(0)$
730	$34(0)$
	0

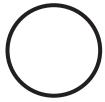
1. What error did Azalea make?
2. What error did Joshua make?

11.2.2 Collaboration: Order of Operations

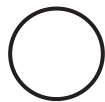
1. Work with your partner to number each operation so that the steps for the order of operations are in the correct sequence.



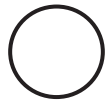
Perform multiplication and division in order from left to right.
(→)



Evaluate expressions with exponents.



Perform addition and subtraction in order from left to right. (→)



Perform operations inside grouping symbols.



An order of operations was first noted in First Year Algebra by Webster Wells and Walter Wilson Hart. The textbook was published in 1912.

2. The problem below is an example from *First Year Algebra*.

EXAMPLE. Find the numerical value of the expression,

$$4ab + \frac{6c}{b} - d^3,$$

when $a = 4, b = 3, c = 5, d = 2.$

- What is the numerical value of the expression? Show your work.
- Check your work with your partner.
- The work shown in the textbook is below.

$$\begin{aligned} 4ab + \frac{6c}{b} - d^3 &= 4 \cdot 4 \cdot 3 + \frac{6 \cdot 5}{3} - 2^3 = 4 \cdot 4 \cdot 3 + \frac{6 \cdot 5}{3} - 2 \cdot 2 \cdot 2 \\ &= 48 + 10 - 8 = 50. \end{aligned}$$

What is the difference between your work and the work from the book?



An order of operations became more important when programmers needed a set of rules to program a computer to find values of numeric expressions. In 1883, Ada Lovelace described in her book about Charles Babbage's analytical engine, the first computer, that codes could be created so the device could handle letters and symbols along with numbers.

11.2.3 Guided Instruction: Evaluating Expressions with Exponents

The work in the book, *First Year Algebra*, showed 2^3 written out as multiplication. This method can get complex as shown in the table.

$(-3)^3 + 2^6 \div -16 + 5$
Exponents written as multiplication
$((-3)(-3)(-3)) + (2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2) \div -16 + 5$

$(-3)^3 + 2^6 \div -16 + 5$
Evaluate expressions with exponents first
$-27 + 64 \div -16 + 5$

- When writing $(-3)^3$ and 2^6 using multiplication, both were written inside a set of parentheses.
- In order of operations, parentheses indicate that the operations inside should be done _____ other steps.
- Therefore, evaluating exponents is done before other multiplication and division when following the order of operations.

1. Evaluate $(-3)^3 + 2^6 \div -16 + 5$.

2. Evaluate each expression, then determine if the expressions are equivalent.

Expression A	Expression B	Equivalent?
$(-3)^4$	$-(3^4)$	
$-(9)^2$	$(-9)^2$	
(-5^2)	$(-5)^2$	
$26 - 4^3$	$26 - (-4)^3$	
$17 + (-2^4)$	$17 - (2^4)$	

3. Describe a strategy that can be used when evaluating expressions with exponents that include a negative sign or a subtraction sign.
4. Elizabeth evaluated $b + a^3 \cdot 7 - 9$ when $a = -4$ and $b = 17$. Her work is below.
- a. Circle Elizabeth's error.

$$17 + (-4)^3 \cdot 7 - 9$$

$$17 + (-64) \cdot 7 - 9$$

$$(-47) \cdot 7 - 9$$

$$-329 - 9$$

$$-338$$

- b. Which step of the order of operations did she not apply correctly?
- A Evaluate expressions with exponents.
 - B Perform operations inside grouping symbols.
 - C Perform multiplication and division in order from left to right.
 - D Perform addition and subtraction in order from left to right.



The order of performing computations is first to work within grouping symbols using the **order of operations**. Then, simplify terms with exponents. Next, while reading from left to right, perform multiplication and division in the order in which it appears. Finally, while reading from left to right, perform addition and subtraction in the order in which it appears.

11.2.4 Try It: Discovering the Importance of Left to Right

1. Mia is evaluating the expression, $x^3 \div r \cdot (k^2 - 5)$ for $k = 3, r = -8$ and $x = 4$. Her first few steps are shown.

$$\begin{aligned}4^3 \div -8 \cdot (3^2 - 5) \\64 \div -8 \cdot (9 - 5) \\64 \div -8 \cdot 4\end{aligned}$$

- a. Mia thinks that she could finish her work by either finding $64 \div -8$ or $-8 \cdot 4$ first. Show the work for both possibilities to determine if Mia is correct.
- b. Was Mia correct when she stated that both methods would work? Use the steps of the order of operations to justify whether or not Mia is correct.
2. Answer the following problems.
- a. Evaluate $r \cdot W \div x \div m$ when $m = 2, r = -4, W = -60$, and $x = -10$. Show your work.
- b. Evaluate $(r \cdot W) \div (x \div m)$ when $m = 2, r = -4, W = -60$, and $x = -10$. Show your work.
- c. How did the use of parentheses change the answers?

a. Evaluate $d \div k \cdot y \div B$ when $B = 3$, $d = -72$, $k = -6$, and $y = 9$. Show your work.

c. How did rearranging the operations change the answers?

11.2.5 Your Turn!

1. Answer the following problems.

a. Circle the expressions you think will result in the same value.

$$7 - (y - 11)^2 + 6V$$

$$7 - y^2 - 11^2 + 6V$$

$$7 - y^2 + 11^2 + 6V$$

b. Find the value of $7 - (y - 11)^2 + 6V$ when $V = -9$ and $y = 4$.

c. Find the value of $7 - y^2 - 11^2 + 6V$ when $V = -9$ and $y = 4$.

d. Find the value of $7 - y^2 + 11^2 + 6V$ when $V = -9$ and $y = 4$.

e. Which expressions, if any, resulted in the same value?

11.2.6 Wrap-Up: Error Analysis

1. Nadine and Nora evaluated the expression $r^3 \div x \cdot c$ for $c = 9, r = -6$ and $x = -3$. Both students made errors in their work shown. Circle their errors.

Nadine’s Work	Nora’s Work
$\begin{array}{l} r^3 \div x \cdot c \\ (-6)^3 \div (-3) \cdot 9 \\ -216 \div (-3) \cdot 9 \\ -216 \div -27 \\ 8 \end{array}$	$\begin{array}{l} r^3 \div x \cdot c \\ (-6)^3 \div (-3) \cdot 9 \\ -18 \div (-3) \cdot 9 \\ 6 \cdot 9 \\ 54 \end{array}$

2. Write a short explanation to each student to explain the mistake and what she needs to remember about evaluating expressions in the future.

To Nadine:	To Nora: